

Chapter 5: Steroids

- **Unit learning outcome:**

- Describe steroids
- Discuss stereochemistry of steroids
- Predict structure along with their IUPAC names of steroidal compounds
- Classify steroids
- Discuss reactions of steroids

- **Contents:**

- 5.1. Introduction (Definition and function)

- 5.2. Stereochemistry of steroids

- 5.3. Nomenclature of steroids

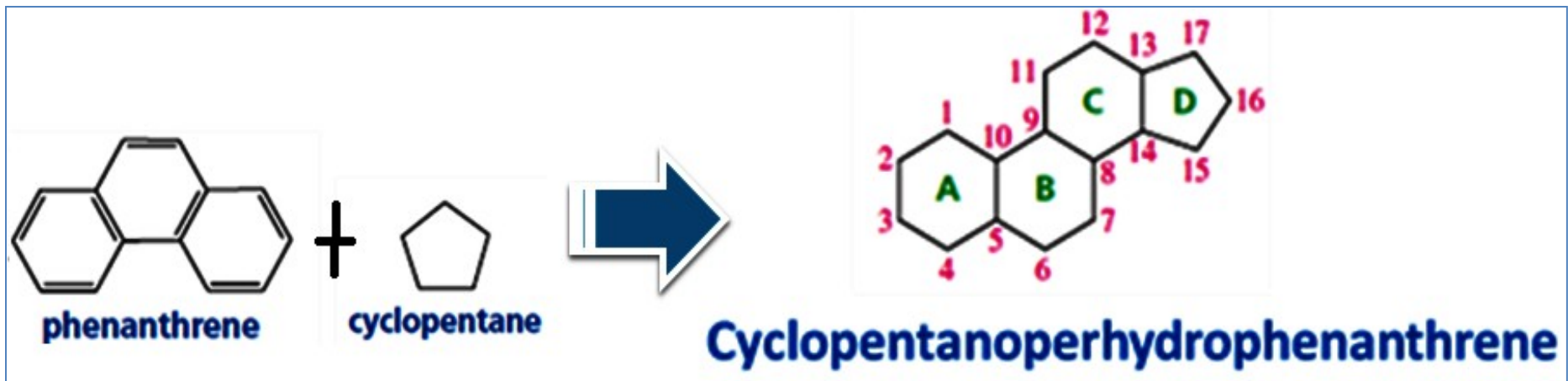
- 5.4. Types of steroids (Anabolic steroids, corticosteroids, sex steroids, sterols, cardiac glycosides, saponins)

- 5.5. Reaction of steroids

5.1. Introduction

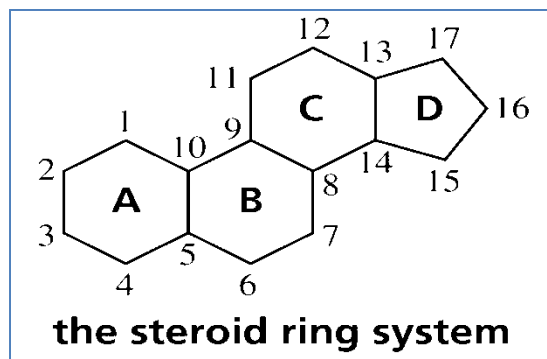
Definition: Steroids

- Naturally occurring compounds and synthetic analogues, based on the cyclopentanoperhydrophenanthrene carbon skeleton
- Natural steroids are derived biogenetically from triterpenoids

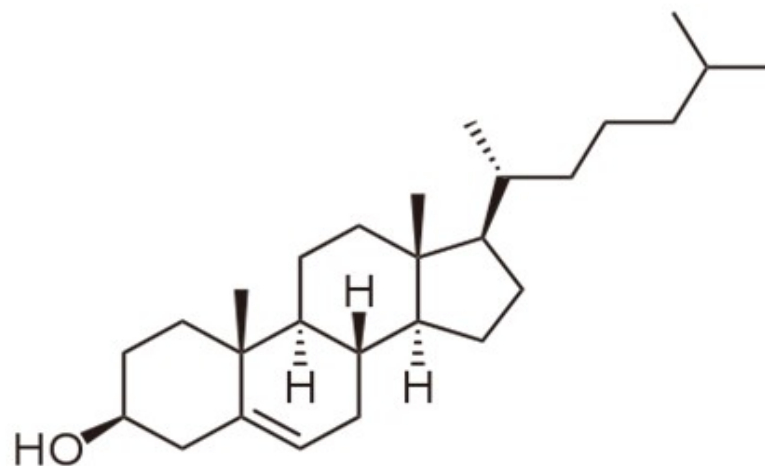


Steroids

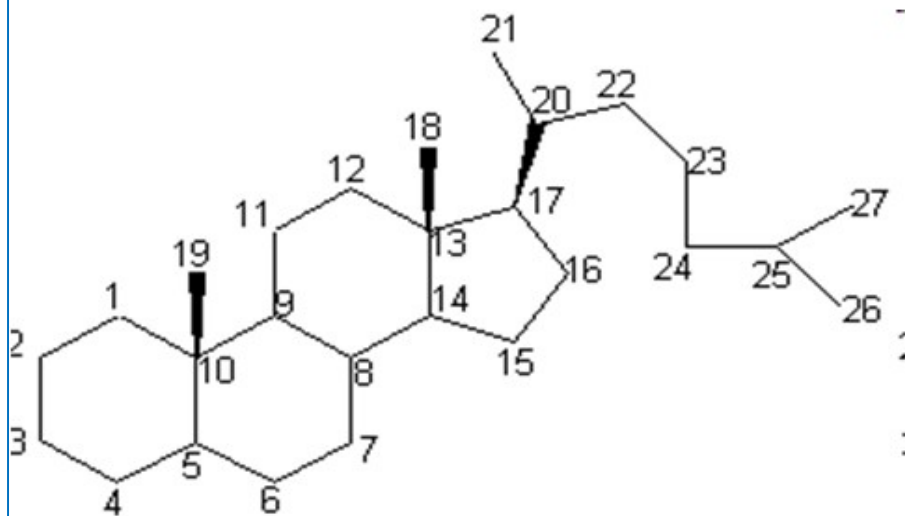
- partially or completely hydrogenated
- there are usually methyl groups at C-10 and C-13 (They are called angular methyl groups),
- and often an alkyl group at C-17.
- by extension, one or more bond scissions, ring expansions and/or ring contractions of the skeleton may have occurred.
- a variety of functional groups may be attached to this basic nucleus at different positions



Numbering



Cholesterol



CHOLESTANE

Significance of Steroids in Pharmacy

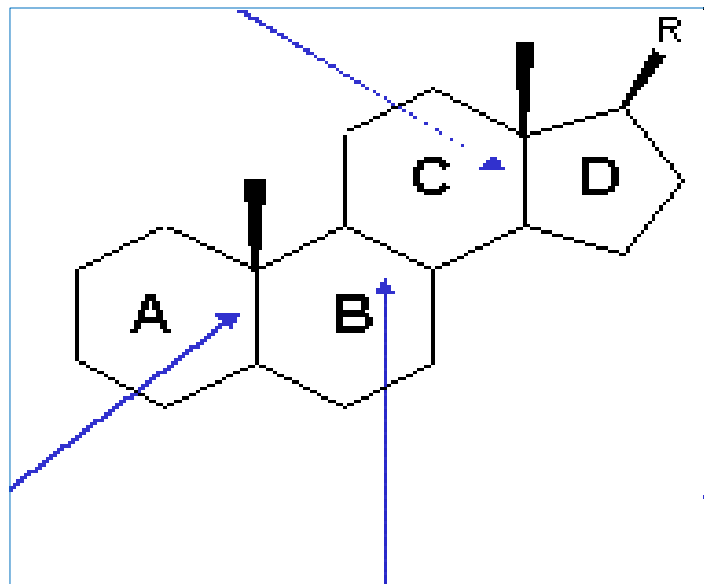
- Steroids play important roles in normal biological processes
 - E.g. Vitamin D, bile salts and sex hormones
- Some steroidal compounds serve as a precursor for the synthesis of sex hormones.
 - E.g. Diosgenin
- Some steroids are either drugs or toxic substances
 - Naturally occurring
 - E.g. Cardiac glycosides
 - Synthetic
 - E.g. Contraceptives, anabolic hormones.....

5.2. Stereochemistry of steroids

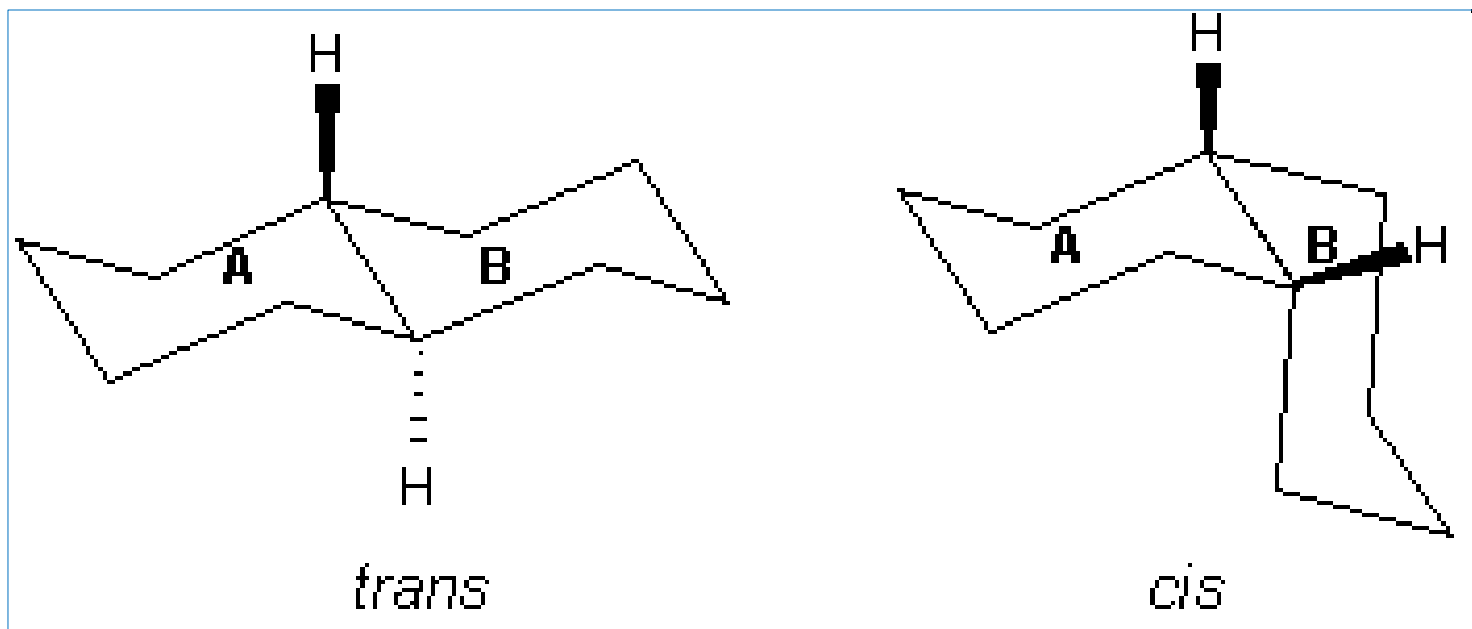
Stereoisomeric sites in steroids

1. *Cis-trans* isomerism at ring junctions

- There are four rings in a steroid skeleton and hence there are three fusion points. A/B, B/C and C/D rings share two carbons each (fusion)

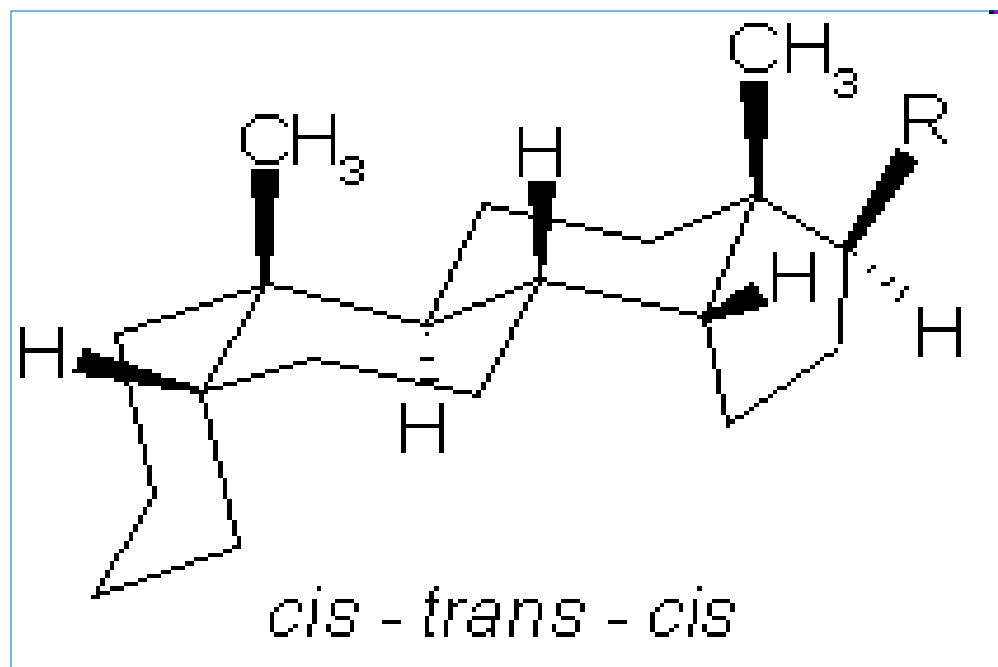
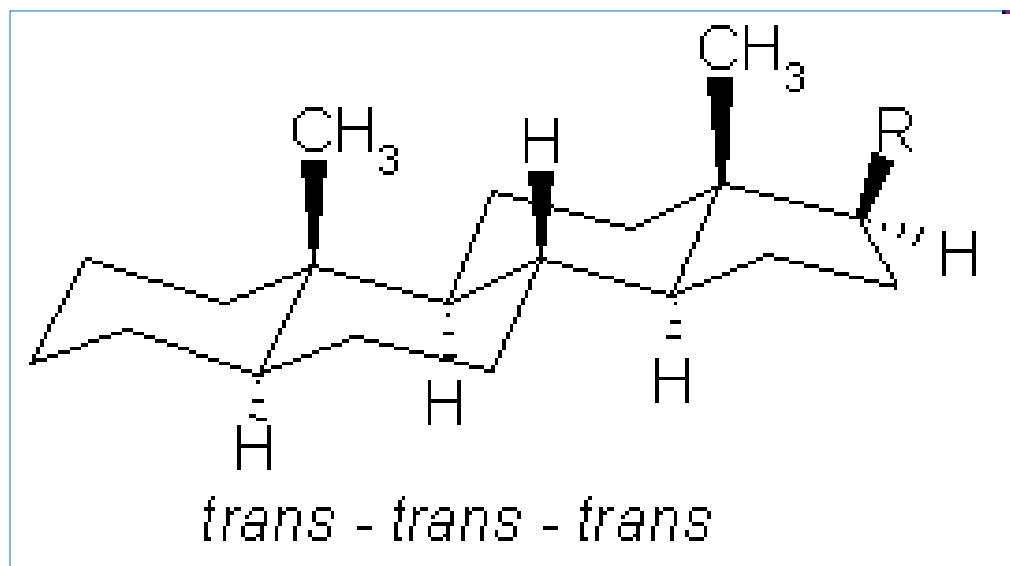


- Every fusion center can either be
- ✓ *cis-* or
 - ✓ *trans*-fused

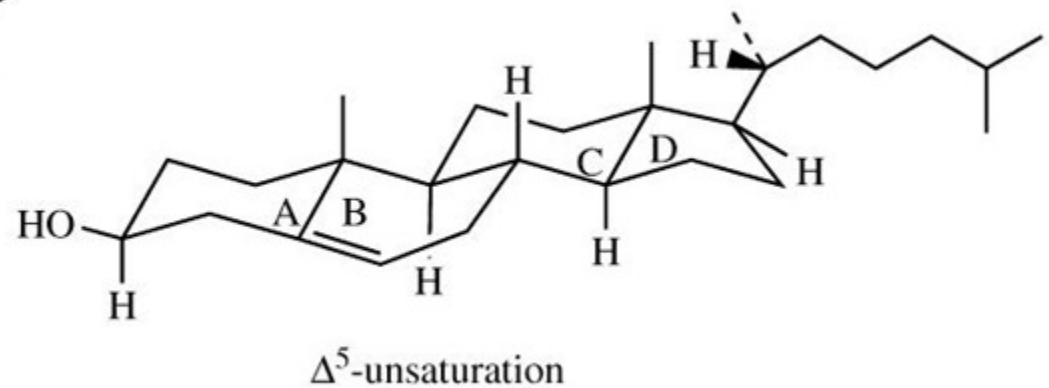
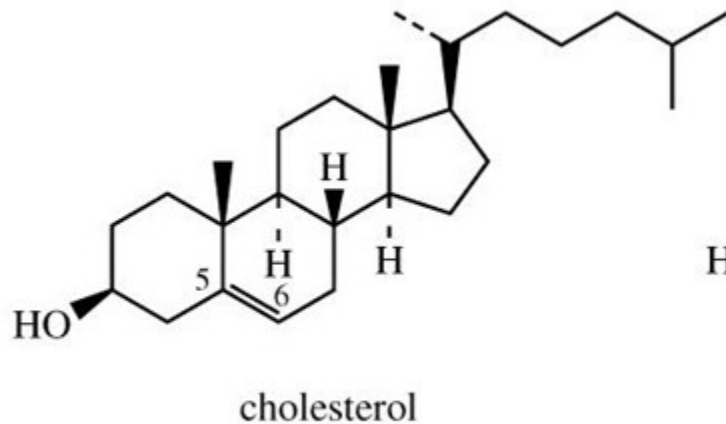
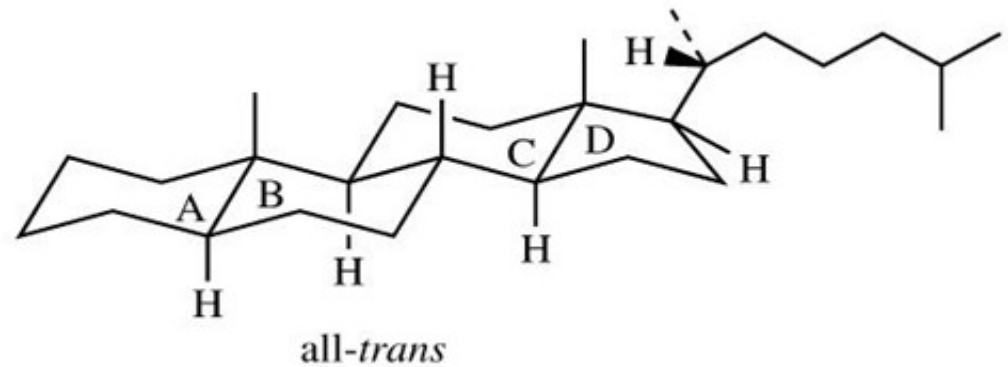
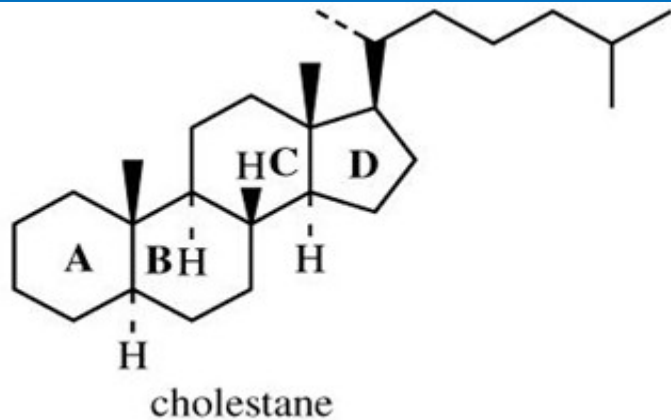


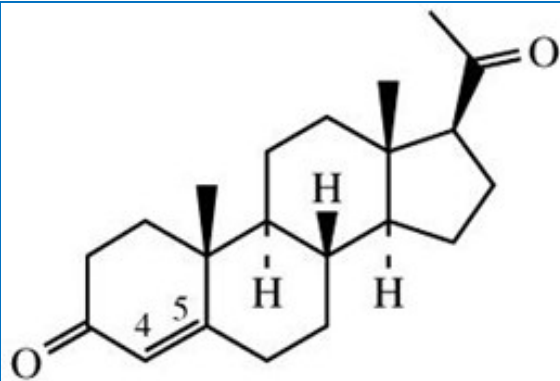
The structures most likely feasible are :

- [trans-trans-trans](#) (most natural and synthetic steroids have this skeleton, e.g., 5 α -dihydrotestosterone)
- [cis-trans-trans](#) (some natural steroids have this skeleton, e.g., cholic acids)
- [cis-trans-cis](#) (few natural steroids have this skeleton, e.g. cardiac glycosides (next semester))
- All natural steroids have a trans B/C fusion, though cis forms can be made synthetically
- In steroids the relation ship between stereochemistry and biological activity is most marked.

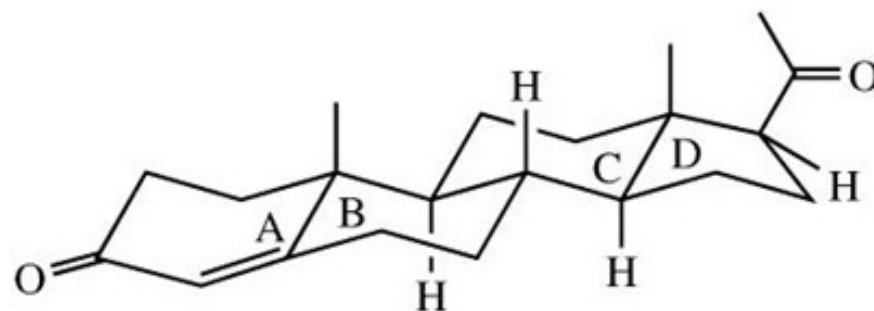


- If double bonds occur at any of rings, **distortion** results

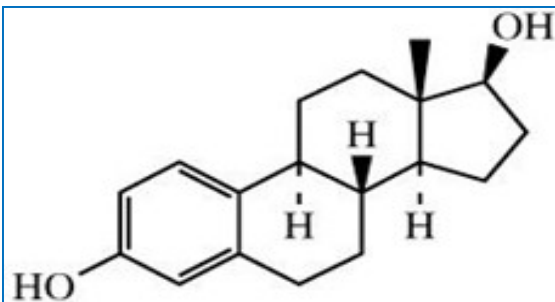




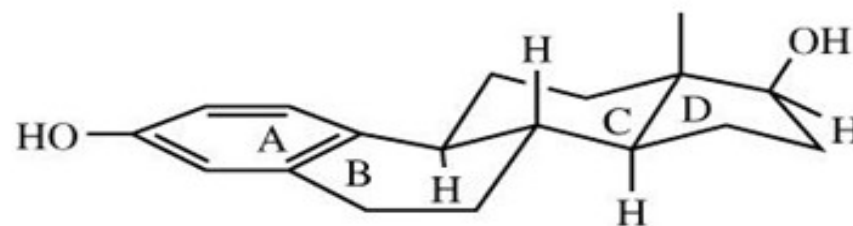
progesterone



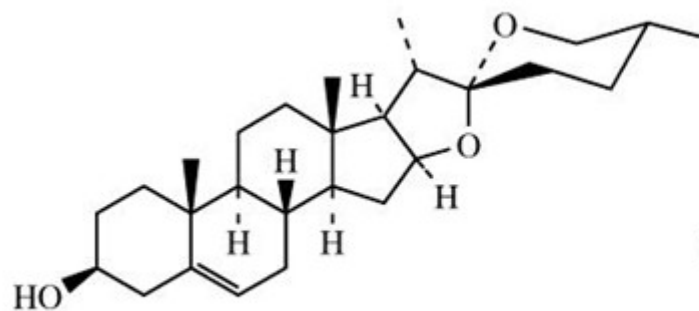
Δ^4 -unsaturation



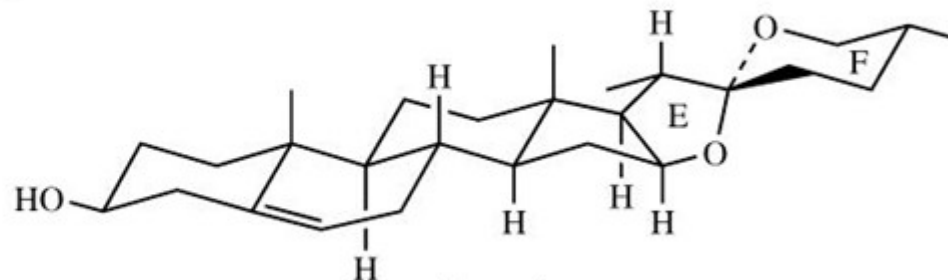
estradiol



A ring aromatic

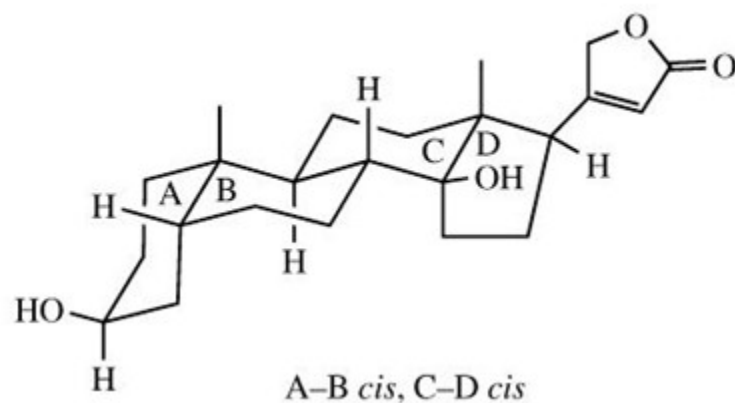
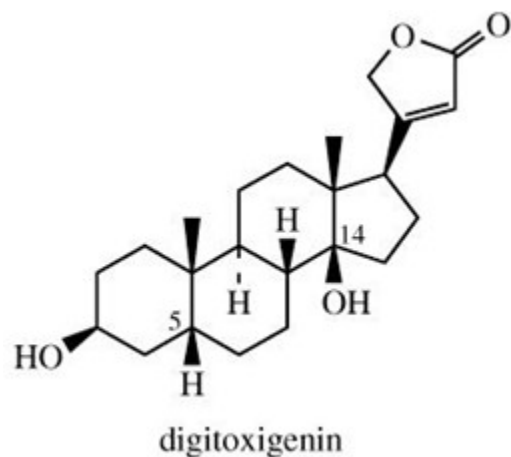
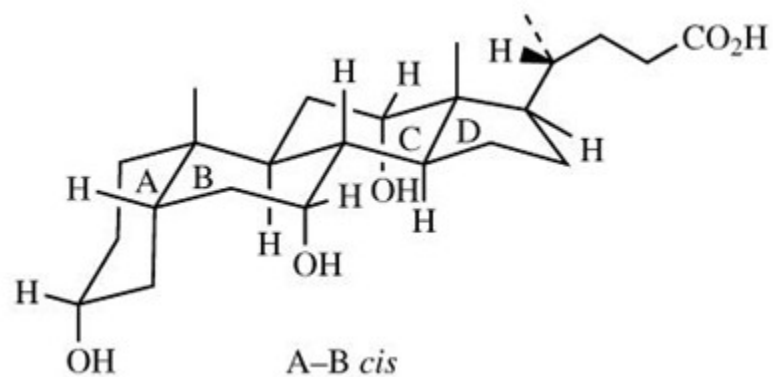
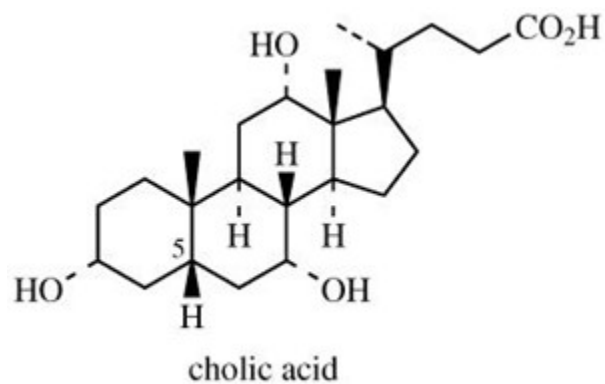


diosgenin

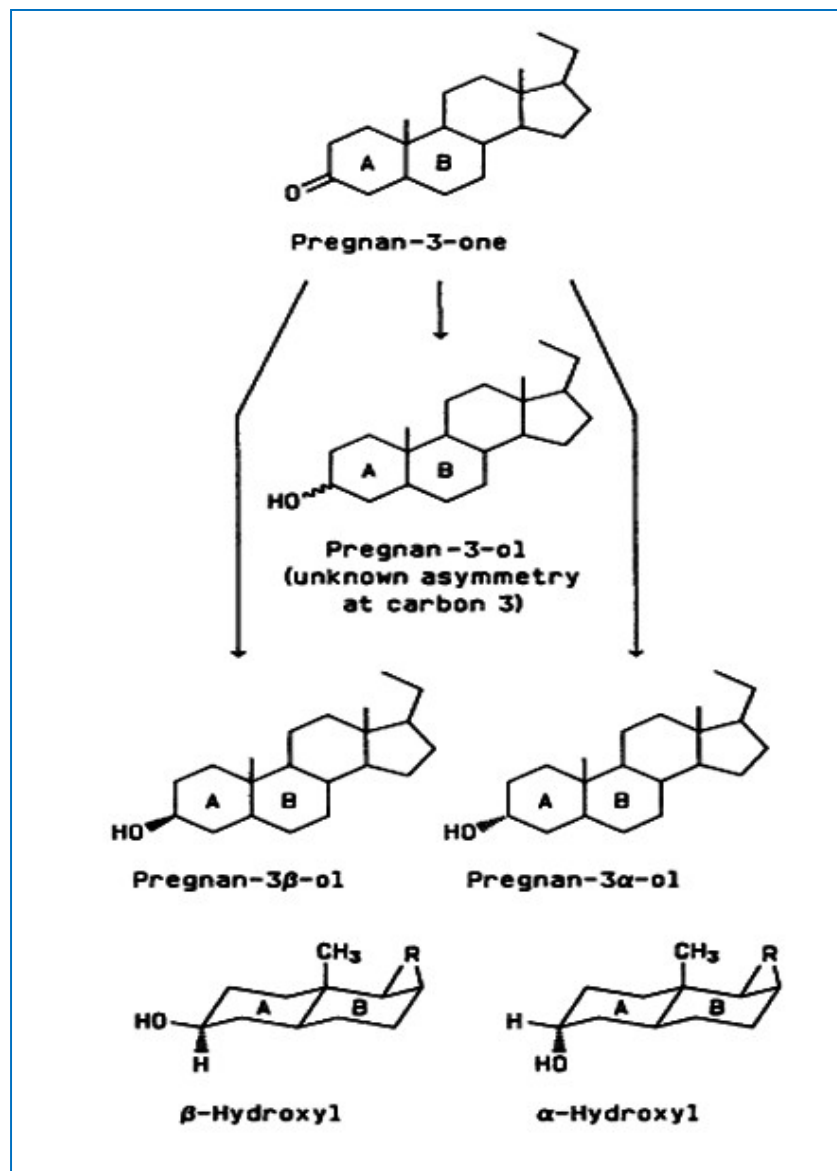


diosgenin

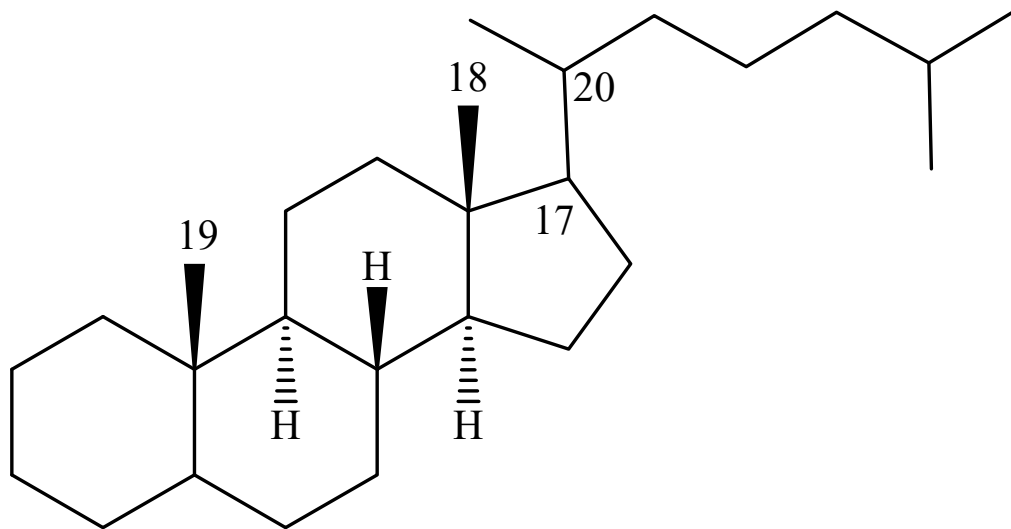
- Examples with cis stereochemistry



2. Asymmetric carbon atoms



3. The *side chain*, usually at C-17



Cholestane (C-27)

5.3. Nomenclature of steroids

- Systematic steroid nomenclature is based on a series of parent saturated hydrocarbon skeletons

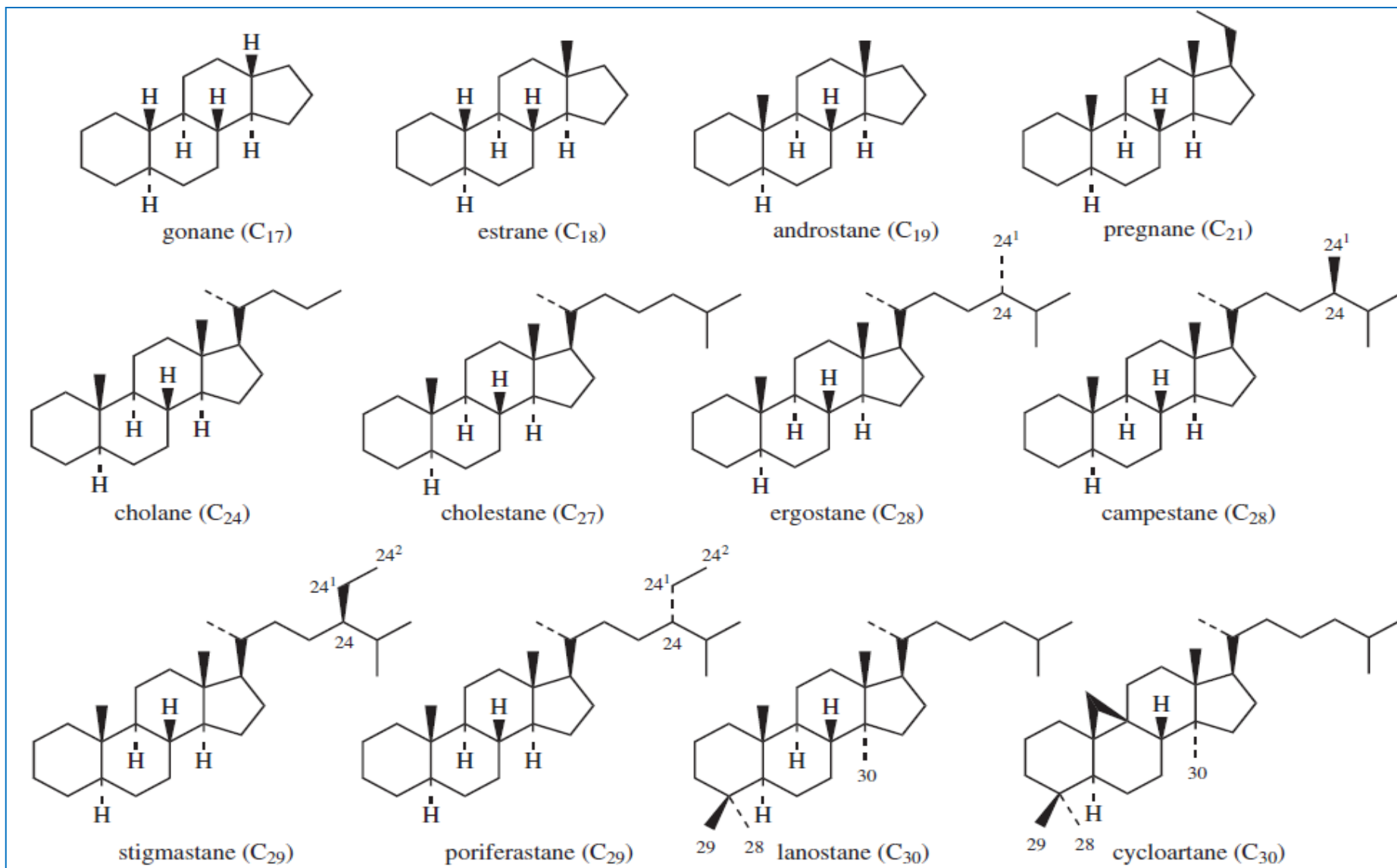
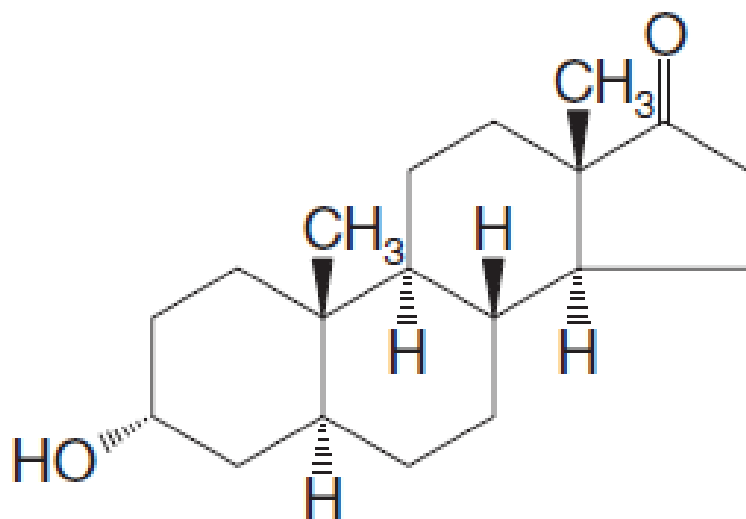


TABLE 1 Steroid Nomenclature Conventions

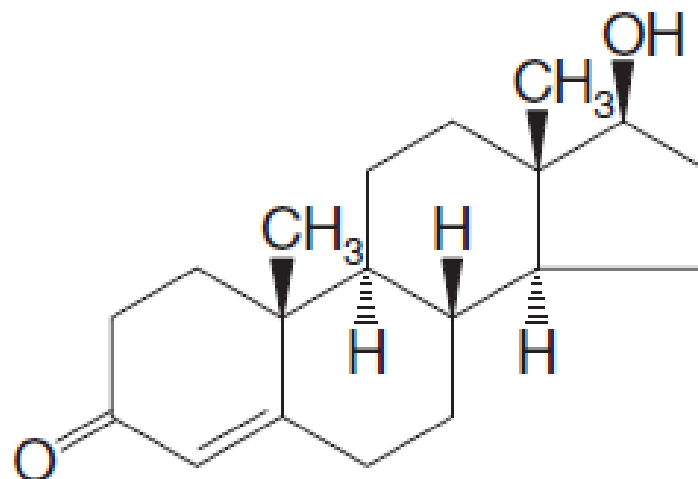
Modification	Prefix	Suffix
Hydroxyl group ($-HO$)	Hydroxy	-ol
Hydroxyl above plane of ring	β -OH	—
Hydroxyl below plane of ring	α -OH	—
Keto or carbonyl group ($C=O$)	Oxo-	-one
Aldehyde ($-CHO$)	—	-al
Carboxylic acid ($COOH$)	Carboxy	-oic acid
Double bond ($-C=C-$)	—	-ene
Triple bond ($-C\equiv C-$)	—	-yne
Saturated ring system	—	-an
One less carbon atom	-Nor	—
One additional carbon atom	-Homo	—
One additional oxygen atom	-Oxo	—
One less oxygen atom	-Deoxy	—
Two additional hydrogen atoms	-Dihydro	—
Two less hydrogen atoms	-Dehydro	—
Two groups on same sides of plane	<i>Cis</i>	—
Two groups on opposite sides of plane	<i>Trans</i>	—
Other ring forms (rings A and B <i>trans</i> , as in allopregnane)	Allo	—
Opening of a ring (as in vitamin D)	Seco-	—
Conversion at a numbered carbon from conventional orientation (as in epicholesterol or 3α -cholesterol)	-Epi	—

- **Exercise 1:**

- Give the systemic name of androsterone and testosterone



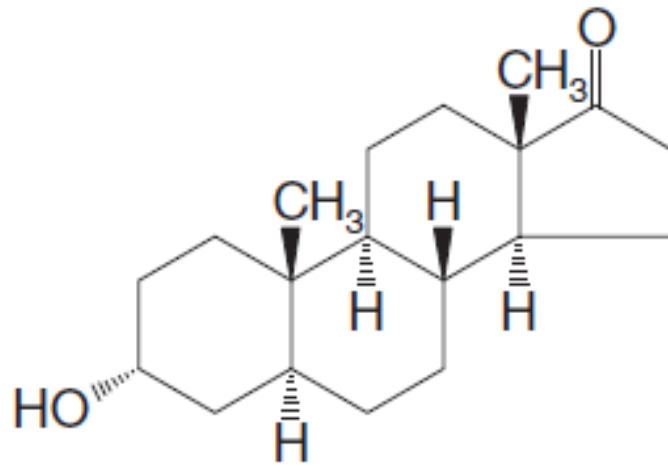
Androsterone



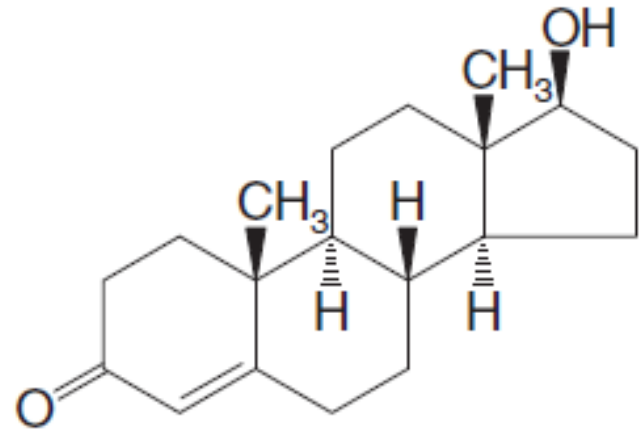
Testosterone

- **Solution 1: for androsterone**
 - Identify the parent group
 - **Androstane**
 - Identify the substituents and their stereochemistry as well as their position
 - 3-hydroxyl: α
 - 17-ketone: Oxo
 - Priority: hydroxyl vs ketone
 - Ketone

- **Answer 1:**

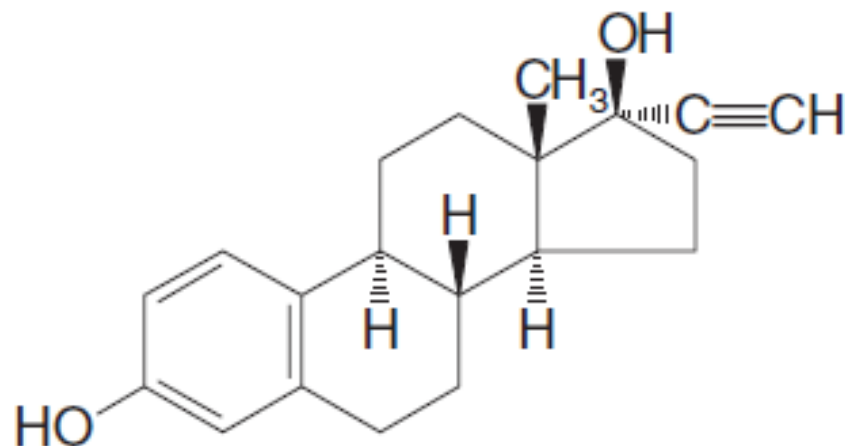


Androsterone
(3 α -hydroxy-5 α -androstan-17-one)



Testosterone
(17 β -hydroxy-4-androsten-3-one)

- **Exercise 2:**
 - Draw the structure of **17 α -ethynyl-1,3,5(10)-estratriene-3,17 β -diol**
- **Answer 2:**



Ethynylestradiol
[17 α -ethynyl-1,3,5(10)-estratriene-3,17 β -diol]

5.4. Physical properties of steroids

- **Steroids are non-polar**
 - High content of hydrocarbons
 - Increases with increase of polar groups such as hydroxyl

5.5. Types of steroids

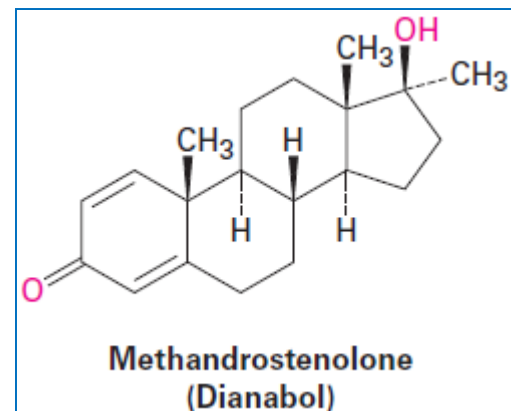
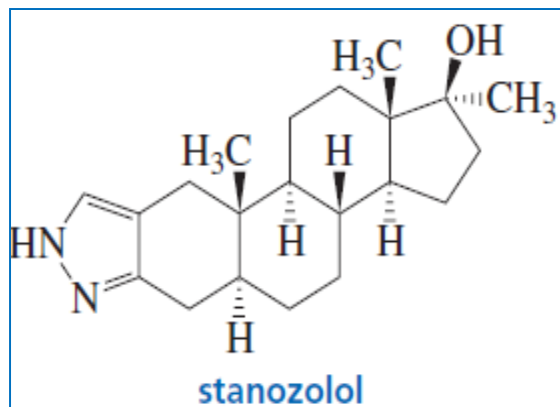
1. Anabolic steroids
2. Corticosteroids
3. Sex steroids
4. Sterols
5. Cardiac glycosides
6. Steroidal saponins

Anabolic Steroids

- Introduced for medical treatment in the 1930's to control tissue breakdown in diseased individuals
- In the 1950's, became popular among athletes to enhance performance
- Became known to increase muscle size and strength at enormous rates
- Injectable by syringe or orally taken
- Many side effects
- Synthesized by humans mainly from the natural form of testosterone

- **Anabolic steroids**

- These agents promote growth of skeletal muscle (anabolic effects) and development of male sexual characteristics (androgenic effects).
- Used in:
 - treating delayed puberty,
 - To treat some types of impotence, and
 - wasting of the body caused by some diseases

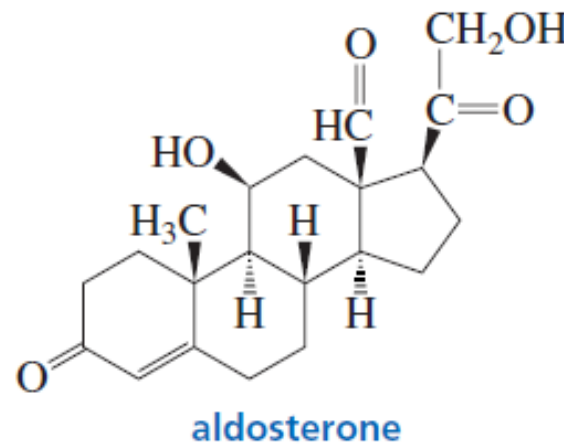
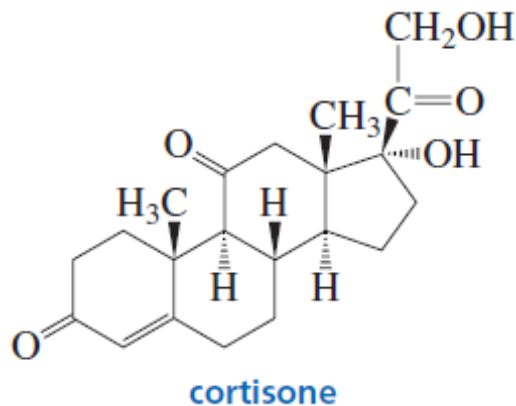


- **Abused:**
 - increase lean body mass,
 - increase strength,
 - increase aggressiveness and
 - lead to a shorter recovery time between workouts



- **Corticosteroids**

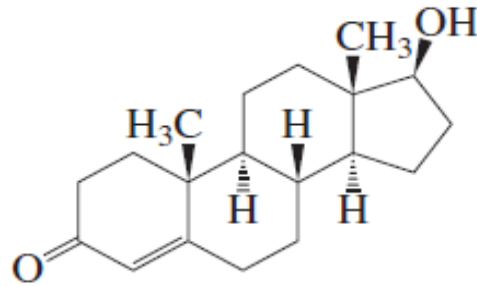
- ✓ are a class of chemicals that includes the steroid hormones that are produced in the adrenal cortex of vertebrates, and synthetic analogues of these hormones
 - Glucocorticoids: such as cortisol control carbohydrate, fat and protein metabolism
 - Mineralocorticoids: such as aldosterone control electrolyte and water levels, mainly by promoting sodium retention in the kidney.



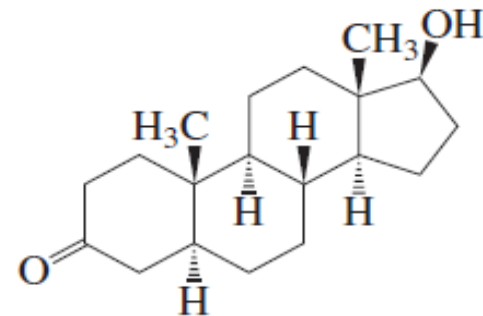
- **Sex steroids**

- ✓ are steroid hormones that interact with vertebrate androgen or estrogen receptors

- Androgens

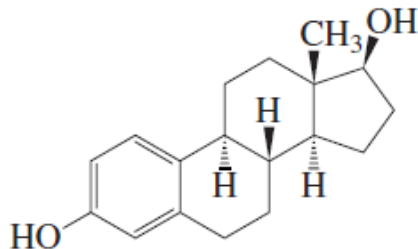


testosterone

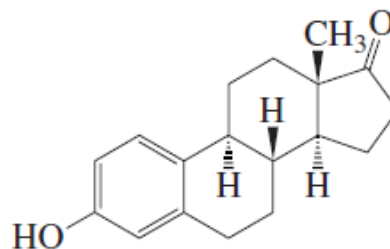


5α-dihydrotestosterone

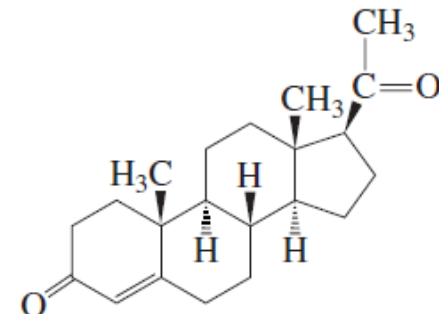
- oestrogens and progestagens



estradiol



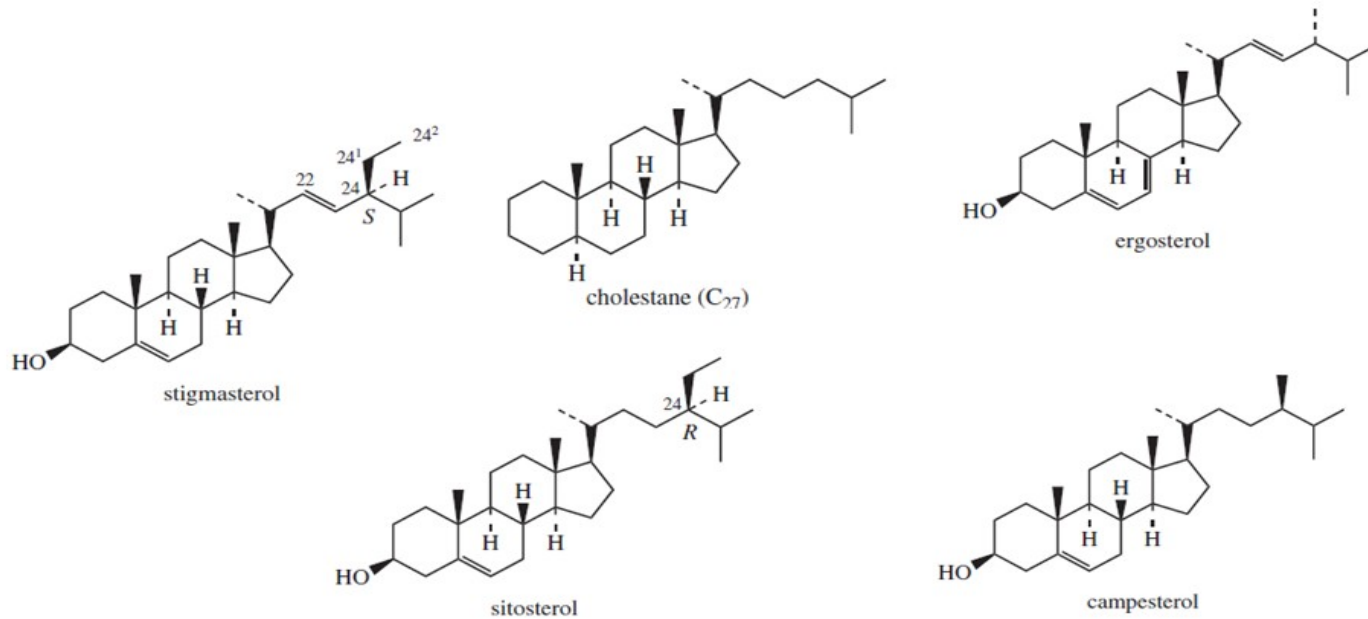
estrone



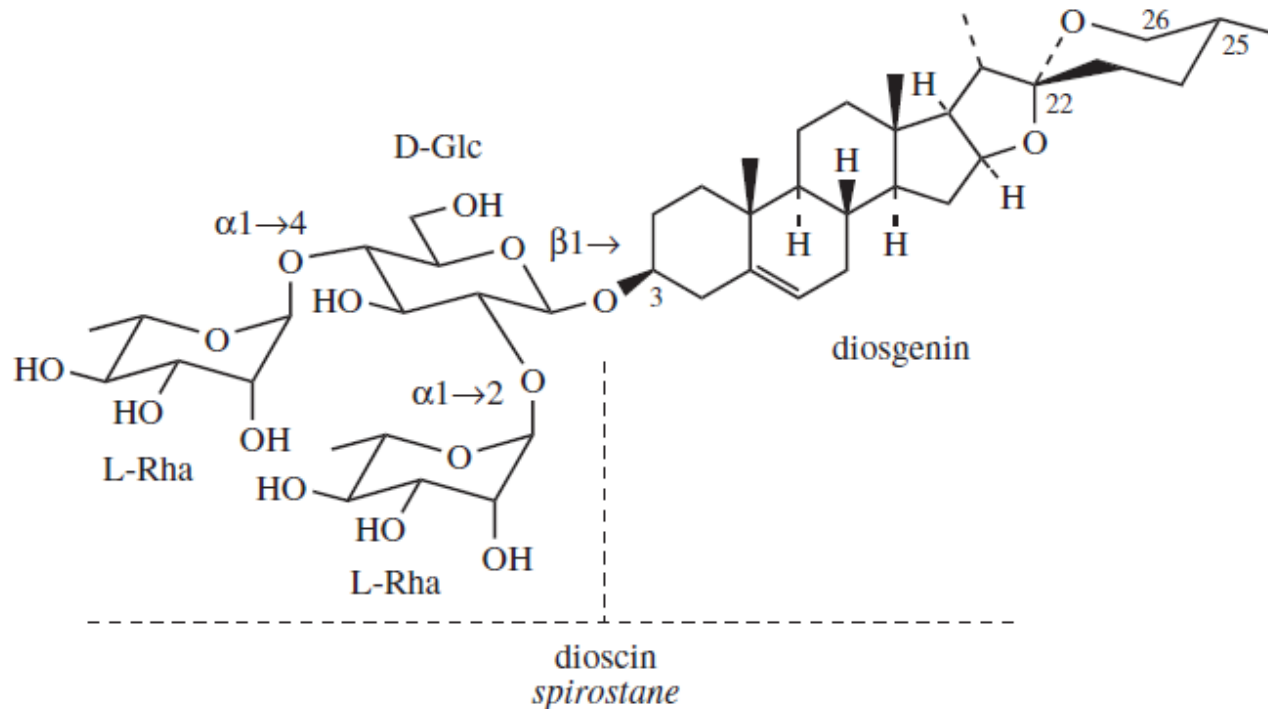
progesterone

Sterols

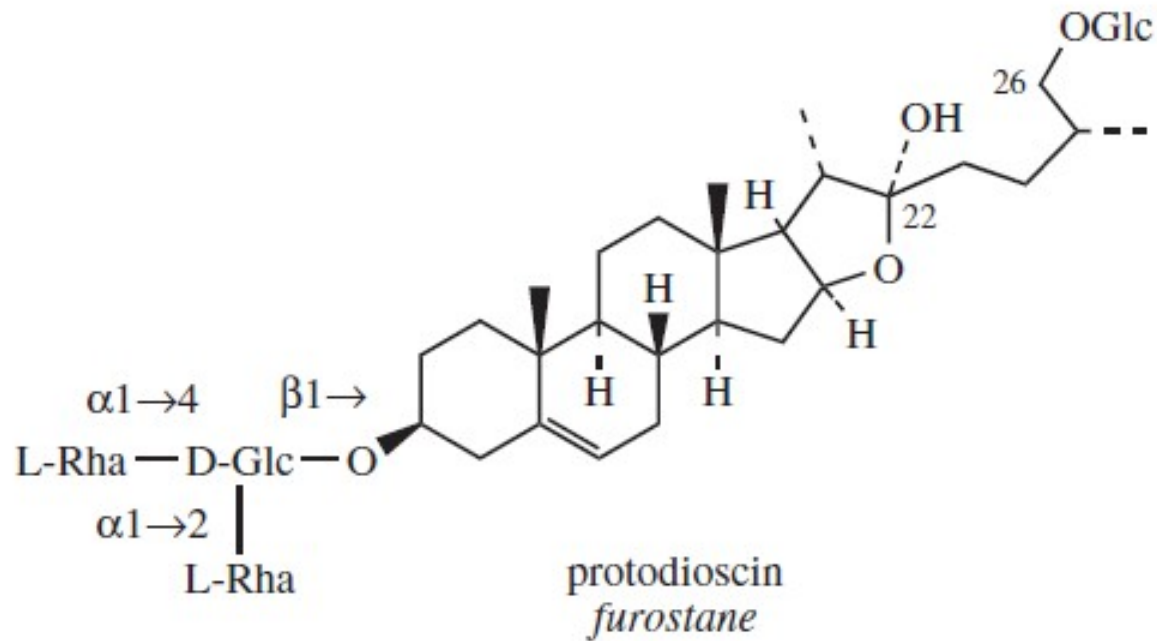
- ☐ also known as **steroid alcohols**, are a subgroup of the steroids and an important class of organic molecules
- ✓ 3-hydroxyl and usually cholestane backbone
 - ✓ Sterols in plants, fungi, and algae



- **Steroidal saponins**
 - Derivatives of cholesterol
 - Two types:
 - (1). Spirostane saponins

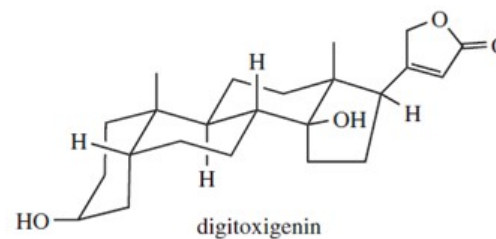
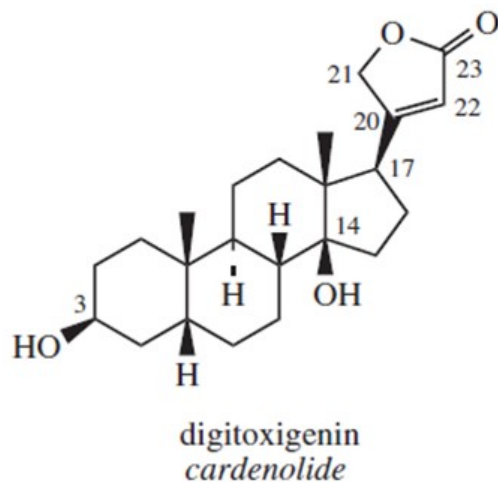
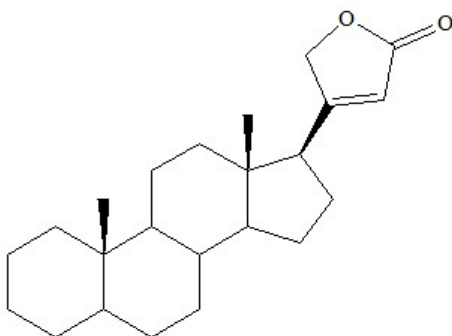


(2). Furostane saponins

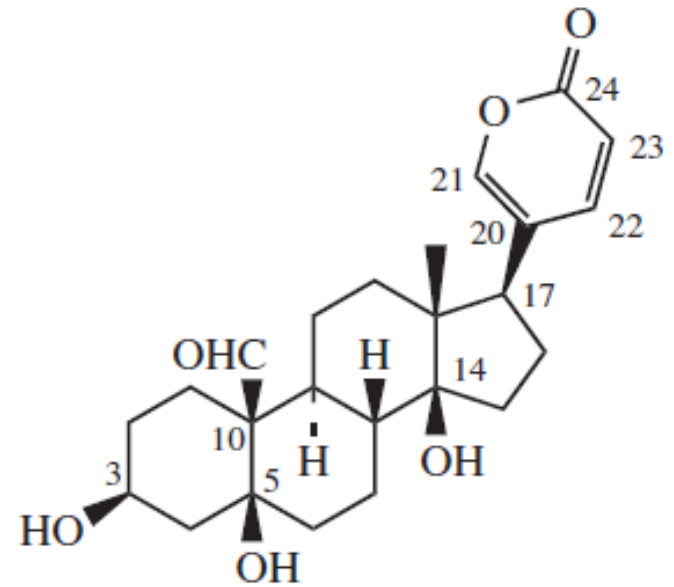
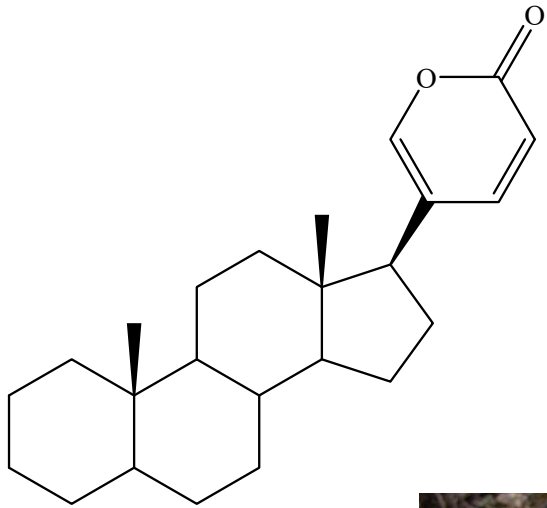


Cardiac glycosides

- Aglycone
 - Cardenolides

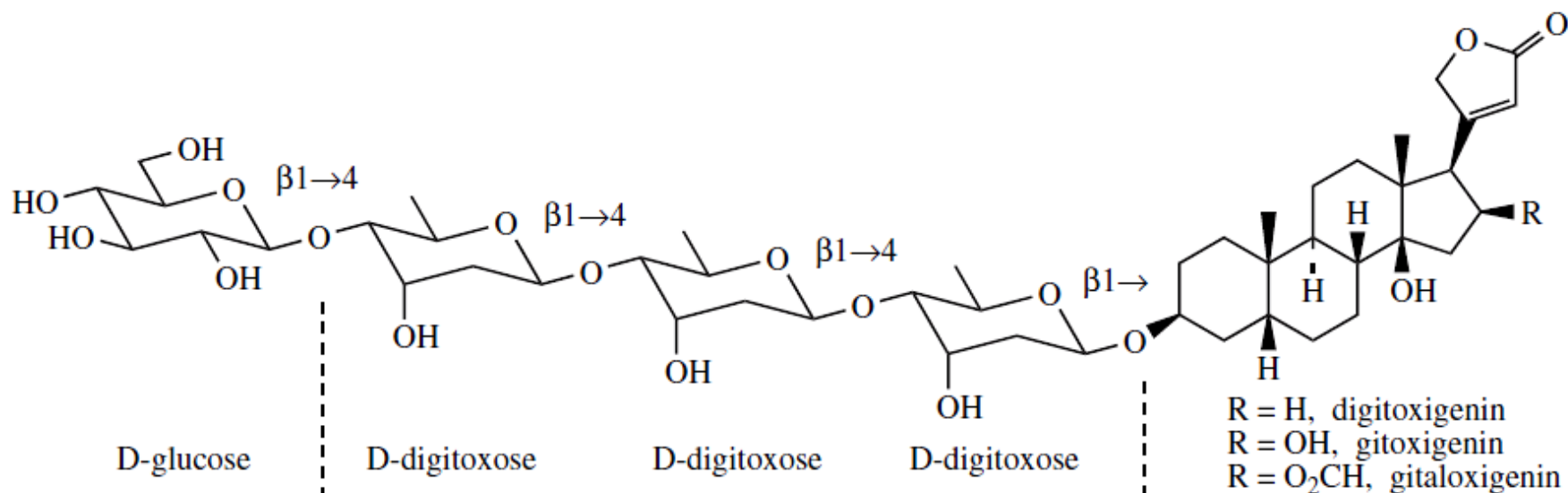
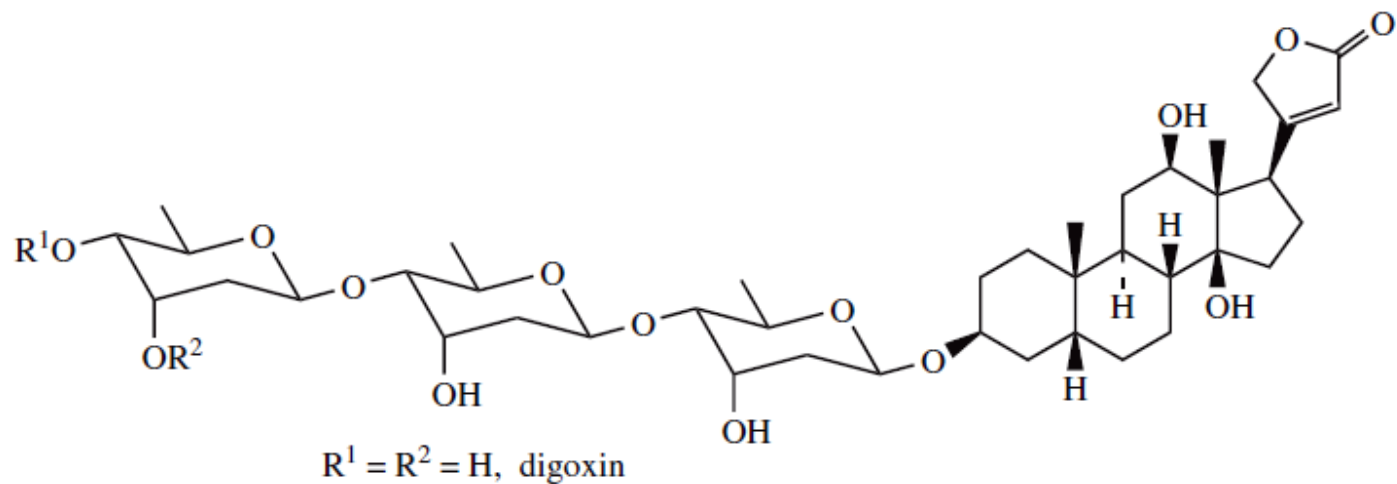


- **Bufadienolides**



hellebrigenin
bufadienolide





$R = H$, digitoxin
 $R = OH$, gitoxin
 $R = O_2CH$, gitaloxin

$R = H$, purpureaglycoside A
 $R = OH$, purpureaglycoside B
 $R = O_2CH$, glucogitaloxin

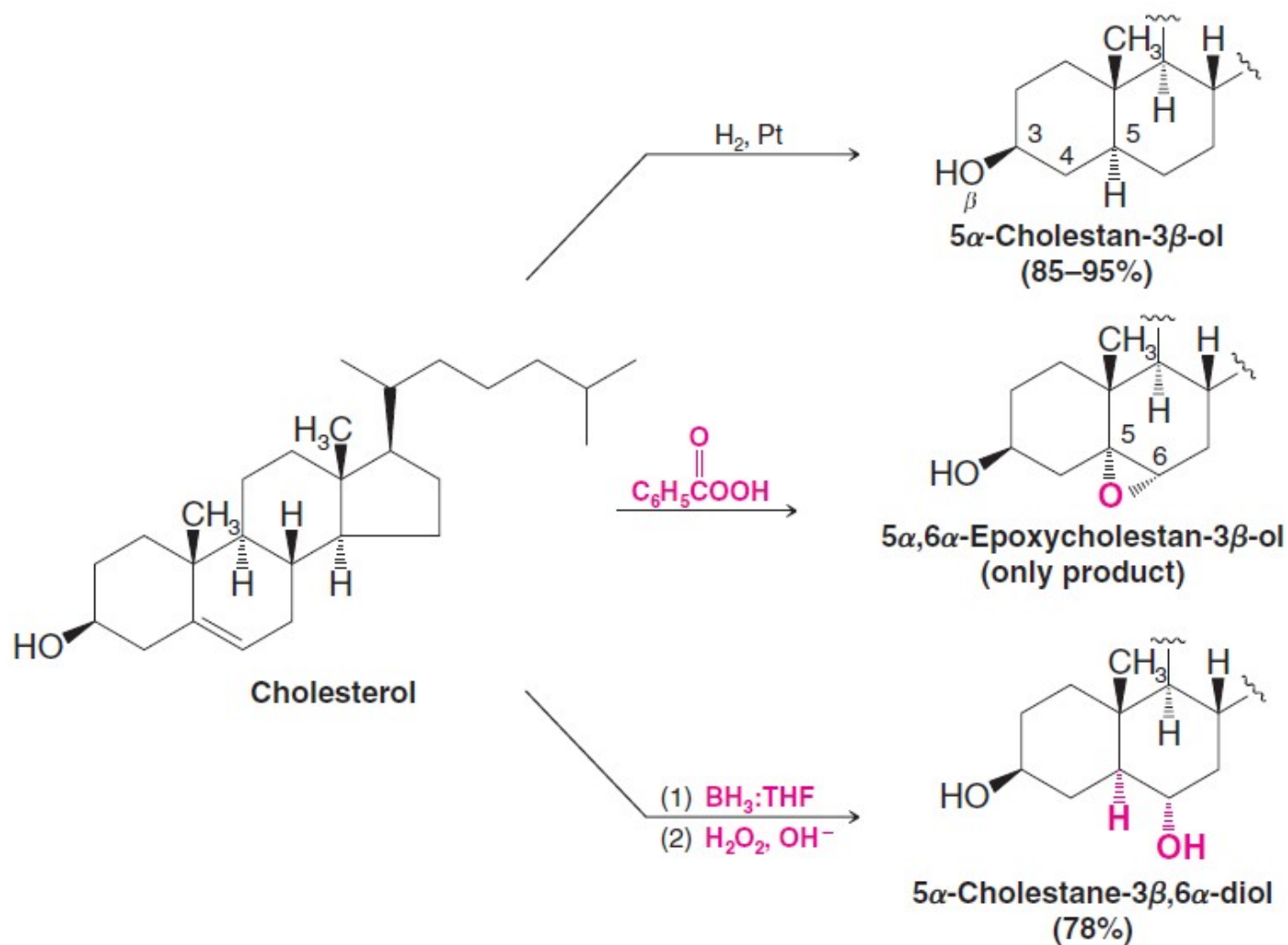
5.5. Biosynthesis of cholesterol

- **Biosynthetic pathway of cholesterol:**

1. the formation of mevalonic acid (six carbons) from three molecules of acetate;
2. $6 \times$ mevalonic acid $\Rightarrow$ squalene (30 carbons);
3. the oxidation and cyclization of squalene into lanosterol
4. the processing of lanosterol to yield cholesterol (27 carbons).
 - This requires a series of 19 steps.

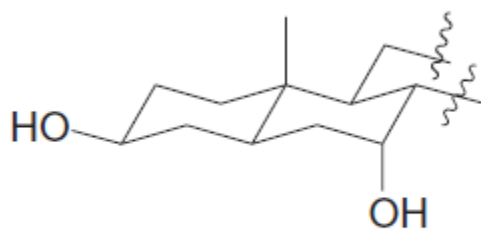
5.6. Reactions of steroids

- **Reaction of steroids depends on:**
 - Functional groups

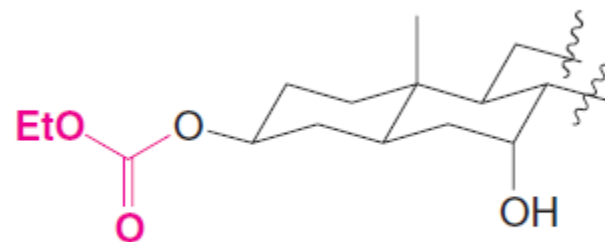
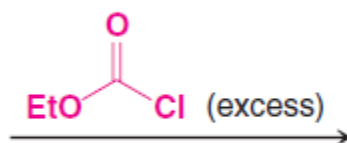


- Influenced by the *steric hindrance*

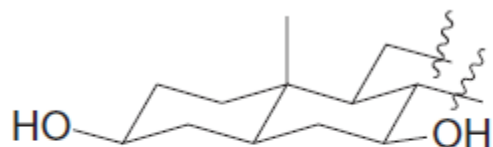
- Esterification of 5 α -cholestane-3 β ,7 α -diol vs 5 α -cholestane-3 β ,7 β -diol



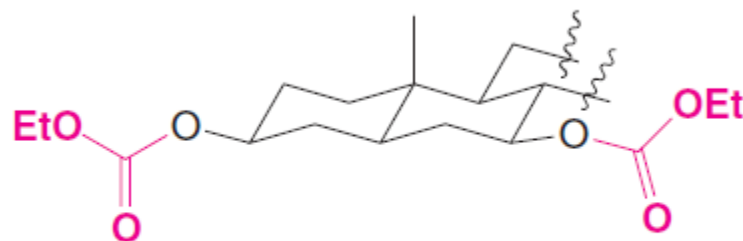
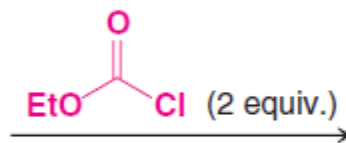
5 α -Cholestane-3 β ,7 α -diol



(only product)



5 α -Cholestane-3 β ,7 β -diol



(only product)

Detection of Sterols

- **The Sakwoski reaction**

- Sterol in $\text{CHCl}_3 + \text{H}_2\text{SO}_4$ Red color

- **The Libermann-Buchard reaction**

- Sterol in $\text{CHCl}_3 + \text{H}_2\text{SO}_4 + \text{acetic anhydride}$

Greenish red color